

MUSTAFA FATHY ABDELTAWAB YOUSSEF

Independent AI Researcher · Microbiology Graduate · Bio-Instrumentation · STEM Educator

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PROFESSIONAL SUMMARY

Microbiology graduate (B.Sc., Al-Azhar University, Faculty of Science — Assiut; May 2026 session; cumulative 77.96%, Very Good) with supervised wet-lab research experience and self-directed expertise in embedded bio-instrumentation. Independently designed two safety-critical control platforms for live biological processes: an ESP32 incubation controller with finite-state safety supervision, predictive sensor-loss fallback, and a quantitative six-factor health score; and an Arduino Mega multi-zone bioreactor for solid-state fermentation.

In parallel, delivered approximately 25–30 hours per week of science and STEM instruction at Al-Matar Official Primary School (classes of 35–50 students), using educational videos, visual materials, and AI tools for lesson design and simplification, while paying close attention to students' psychological well-being. Also provided private science tutoring for primary grades 4 and 5 for approximately two years and foundational English teaching.

From secondary school through university, fully self-financed personal education through consistent, multi-year part-time and seasonal work. Built practical leadership, operational reliability, and customer-facing skills first as an agricultural greenhouse contractor and labor supervisor (three years), then through progressive retail and hypermarket roles culminating in full operational responsibility during peak seasons.

On 11 September 2026, published the independent research preprint HDAFA (Hyper-Deterministic Atomic Agentic Architecture for Verifiable Code Generation) on Preprints.org (manuscript ID 232818) with a Zenodo DOI (10.5281/ZENODO.22700296), open-sourcing the full code, benchmark scripts, and experimental data.

LATEST ACHIEVEMENT — 11 SEPTEMBER 2026

HDAFA Preprint Publication

Authored and released HDAFA: A Hyper-Deterministic Atomic Agentic Architecture for Verifiable Code Generation as an independent research preprint.

- Preprints.org manuscript ID: 232818
- Zenodo DOI: 10.5281/ZENODO.22700296
- Preprint URL: <https://www.preprints.org/manuscript/202609.232818>
- Code repository: <https://github.com/MostafaFathi-afroto/hdafa>
- ORCID record: <https://orcid.org/0009-0000-4920-9333>
- Submitted: 11 September 2026
- Status: Under initial review (Preprints.org)
- Subject Area: Computer Science → Artificial Intelligence and Machine Learning
- License / openness: Open-source code (MIT), supplementary data archived on Zenodo

Scientific Content of the Contribution

- Designed a three-agent atomic pipeline: Requirements Analyzer → HTML/CSS Coder → Deterministic Validator.
- Built a zero-LLM deterministic validator with 10–11 algorithmic structural checks (DOCTYPE, lang="ar", dir="rtl", style tag, responsive signals, Arabic text presence, tag/div balance, title, viewport, body, no JavaScript).
- Defined a keyword-based section-coverage metric to measure content completeness beyond structural pass rate.
- Ran a controlled comparison of four systems under a fixed model (gemini-3.5-flash-lite, temperature=0.0): Single-Agent, Single+Validator, Multi-No-Validator, and HDAFA.

- Evaluated 14 completed web-generation tasks across five categories (services, e-commerce, landing pages, personal portfolios, edge cases).
- Primary empirical result: Analyzer-based systems achieved 100% average section coverage versus approximately 85% for direct-generation baselines — a 15 percentage-point improvement.
- Largest gap observed on a hard edge-case task (12 sections): Single-Agent coverage 50% vs Analyzer-based systems 100%.
- HDAFA average retries on completed tasks: 0.00 — documenting that the validator repair loop was not activated under this task set.
- One additional hard-constraint task failed during Analyzer schema generation and was reported as a failure case (not silently omitted).
- Released full reproducibility package: source code, prompts, benchmark runner, raw CSVs, summary tables, chart, and sample HTML outputs.

Scientific Integrity Notes Documented in the Paper

- Explicitly stated that the main measured gain is attributable to the Requirements Analyzer, not to validator-guided recovery in this benchmark.
- Acknowledged coverage-measurement threat if reference schemas are not independent (gold-schema protocol recommended for future work).
- Acknowledged small sample size, single-run protocol, HTML/CSS-only scope, and model dependence.
- Framed the work as an early empirical technical report / preprint, not as a final large-scale conference claim.

PUBLICATIONS

Youssef, M. F. A. (2026). HDAFA: A Hyper-Deterministic Atomic Agentic Architecture for Verifiable Code Generation. Preprints.org.

- Preprints ID: 232818
- DOI (Zenodo): <https://doi.org/10.5281/ZENODO.22700296>
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- Code: <https://github.com/MostafaFathi-afroto/hdafa>
- Submitted: 11 September 2026 | Status: Under initial review

RESEARCH PROJECT — HDAFA

HDAFA — Hyper-Deterministic Atomic Agentic Architecture

Independent Researcher · September 2026 – Present · <https://github.com/MostafaFathi-afroto/hdafa>

Designed and implemented a multi-agent system that wraps probabilistic LLMs inside a deterministic verification shell to produce verifiable HTML/CSS web pages, with particular attention to Arabic (RTL) contexts.

- Architecture: 3-agent pipeline — Requirements Analyzer (natural language → JSON schema), HTML/CSS Coder (schema → page), Deterministic Validator (algorithmic PASS/FAIL + coverage).
- Deterministic Validator: zero AI involvement; structural gate + section-coverage gate.
- Benchmark protocol: 14 tasks × 4 systems × fixed model and temperature.
- Results: +15 pp average coverage from structured requirements analysis; HDAFA matched Multi-No-Validator on primary quality metrics in this set.
- Reproducibility: open-sourced agents, benchmark scripts, raw results, and analysis outputs.
- Tools: Python 3.11+, Google Gemini API (gemini-3.5-flash-lite), Pandas, Matplotlib, Git, GitHub, Google Colab.

ENGINEERING & RESEARCH PROJECTS

ESP32 Safety-Critical Bio-Incubator Controller (v2.0.1)

Independent Project · 2025–2026 · github.com/MostafaFathi-afrotoh/ESP32-Egg-Incubator

Solved the gap between basic thermostat code and long-running biological control by shipping offline-first ESP32 firmware for precision incubation (37.7 °C target; 37.4–37.8 °C hysteresis; 45–55% RH via evaporator).

- Reduced the risk of an energized heater after a software fault by implementing a four-state safety supervisor (BOOT / NORMAL / SAFE / EMERGENCY) with a hardware-level heater cut independent of high-level state logic.
- Addressed total sensor-loss by building a predictive thermal fallback from learned heat/cool rates, with confidence decaying at 0.5%/min and automatic Safe Mode entry below 60% confidence or outside the physiological safety band.
- Made system condition measurable with a 0–100 Health Score from six weighted factors and a 32-entry severity-tagged fault ring.
- Improved remote diagnosis via Telegram diagnostic commands and Adafruit IO feeds over non-blocking Wi-Fi with exponential backoff.
- Root-caused and fixed unexplained reboot loops after Arduino-ESP32 core 3.x migration.

Automated Bioreactor Master Controller (SystemUnite4)

Academic Research Project · 2025–2026 · github.com/MostafaFathi-afrotoh/Automated-Bioreactor

Engineering supervision: Prof. Dr. Salah Badr, Dr. Ragab Qasem; Biological practice under Dr. Omar Atta

- Programmed Arduino Mega 2560 firmware with dual DS18B20 zones, dual heater relays, dual aeration, and I2C LCD status.
- Implemented HC-05 Bluetooth protocol for remote process control and a Kivy/KivyMD monitoring interface with multi-threaded sensor streaming.
- Supported laboratory reproducibility with architecture/pinout documentation and post-run analysis in Pandas and Matplotlib.

Hormone Quantification Methods & Nanobiosensors — Graduation Project

Supervised by Dr. Omar Atta

- Compared immunoassays versus nanobiosensors for hormone detection: analytical sensitivity, limit of detection, and signal transduction at nanostructure interfaces.

Biocontrol of Fusarium & Actinomycetes Screening

Microbiology Laboratory Project under Dr. Omar Atta

- Investigated biocontrol strategies against *Fusarium* sp. using plant extracts with coatings.
- Built culture-handling competence through isolation, purification, and antimicrobial screening of *Actinomycetes* isolates.

WET-LAB COMPETENCIES (SUPERVISED RESEARCH)

- Aseptic handling, routine microbial culture, isolation, and sterile workflow (media work, contamination control, sample labeling)
- *Actinomycetes* isolation, purification, and antimicrobial screening
- Solid-state fermentation (SSF) process sampling
- Standard laboratory equipment: incubators, laminar flow cabinet, hot-air oven, colony counter, pH meter, spectrophotometer, centrifuge, fermentor, rotary evaporator, PCR, paper chromatography
- Biosafety-aware handling, instrument hygiene, and reproducible record-keeping

EDUCATION

Bachelor of Science (B.Sc.) in Microbiology

Al-Azhar University — Faculty of Science for Men, Assiut

May 2026 session | Cumulative: 5,925 / 7,600 marks (77.96%) — Grade: Very Good

Faculty Council: 14 July 2026 | University Council: 21 July 2026

- Relevant coursework: Microbial Physiology, Biochemistry, Microbial Genetics, Cytology, Mycology, Immunology, Microbial Taxonomy, Microbial Biotechnology, Applied Microbiology, Genetic Engineering
- Final-year research under Dr. Omar Atta: nanobiosensors and hormone quantification methods

TEACHING EXPERIENCE

Volunteer Science Educator — Al-Matar Official Primary School, Alexandria

February – May 2026 | Approximately 25–30 hours per week | Official Experience Certificate issued

- Delivered science and STEM instruction for grades 1–6, managing large classes of 35–50 students and developing strong classroom leadership and communication skills.
- Used educational videos, YouTube content, and visual materials extensively to make scientific concepts clear and memorable.
- Simplified complex scientific ideas into easy, age-appropriate explanations tailored to different student levels.
- Leveraged AI tools heavily to design and simplify lessons, create teaching materials, and support lesson delivery.
- Paid close attention to students' psychological well-being, creating a supportive classroom atmosphere that encouraged participation and confidence.

Private Science Teacher (Primary) — Alexandria

During university studies | Approximately 2 years (part-time)

- Taught Science to primary students in grades 4 and 5, focusing on clear explanations, visual learning, and building foundational scientific understanding.
- Designed simplified lessons that matched students' comprehension levels.

Private English (Foundation) Tutor — Alexandria

3 months

- Taught foundational English to primary learners, integrating basic literacy with simple STEM-related vocabulary and concepts.

Private English & STEM Tutor — Alexandria

July 2026 – Present

- Continues delivering structured English literacy and STEM problem-solving sessions for primary learners.

RETAIL, SALES & HYPERMARKET OPERATIONS EXPERIENCE

Cashier, Storekeeper & Hypermarket Supervisor (Seasonal)

El-Salheya Hypermarket for Protected Agriculture · Casa Bianca village, after El-Hammam Gate, Marsa Matrouh

Three consecutive summer seasons during university studies

- Progressed from sales-floor associate to cashier, then to storekeeper, and ultimately to in-charge / supervisor of hypermarket operations during peak vacation months — reflecting demonstrated reliability and repeated trust from management over three consecutive seasons.
- Managed point-of-sale cashier operations accurately and efficiently during high-footfall summer periods.
- Handled goods receiving and dispatch (stock intake and hand-over), supporting inventory accuracy on a high-turnover retail floor.
- Directly supported customers on the sales floor: product guidance, queries, and point-of-sale service under time pressure.
- Assumed full operational responsibility for hypermarket floor operations during vacation months, coordinating daily tasks independently and acting as overall supervisor.
- Dealt with suppliers, sellers, companies, and representatives; gained practical understanding of sales, purchasing, inventory flow, and day-to-day retail management.
- Developed core commercial skills: accurate cash handling, customer service under pressure, stock control, supplier coordination, and progressive operational leadership.

AGRICULTURAL GREENHOUSE CONTRACTING & SUPERVISION

Agricultural Greenhouse Contractor & Labor Supervisor

Mohamed Naguib Agricultural Area · Three years during secondary school (approx. 3rd preparatory through 3rd secondary)

Typically 2–3 days per week; full-time during school vacations

- Worked as a contractor and supervisor of own labor team in agricultural greenhouses, taking direct responsibility for work execution and team coordination over a multi-year period.
- Self-financed secondary school studies through this sustained part-time work, developing early financial independence, discipline, and personal accountability.
- Gained practical leadership and supervisory skills by managing labor under real field conditions in a remote, highly disciplined agricultural environment.
- Acquired hands-on knowledge of smart greenhouse systems (Spanish smart greenhouses, developed Spanish types, Egyptian systems, and related protected cultivation structures).
- Learned practical plant cultivation methods, seasonal cycles, correct plant rearing practices, and field-level prevention and protection techniques over an extended period.
- Built foundational habits of reliability, punctuality, physical endurance, and self-reliance while balancing structured part-time work with academic responsibilities.

TECHNICAL SKILLS

- **AI / LLM:** Google Gemini API, Prompt Engineering, Structured Output Design, Temperature Control (0.0), Multi-Agent Orchestration, Deterministic Verification Design.
- **Data & Software:** Python 3.11+, Pandas, NumPy, Matplotlib, CSV/JSON pipelines, Benchmark Design, Kivy/KivyMD.
- **Verification:** Deterministic Checks, Regex Validation, Structural Verification, Section Coverage Metrics, JSON Schema Validation.
- **Embedded Hardware & Firmware:** ESP32, Arduino (Uno/Mega), C/C++ (OOP, finite state machines), MicroPython, FreeRTOS basics.
- **Control, Safety & Reliability:** Closed-loop hysteresis control, EMA filtering, predictive fallback, fault logging, weighted health scoring.
- **Connectivity & Monitoring:** Non-blocking Wi-Fi, Telegram Bot API, Adafruit IO telemetry, Bluetooth Serial (HC-05).
- **Hardware Design:** Schematic/PCB layout (EasyEDA, Fritzing), 3D enclosure concepts (FreeCAD).
- **Laboratory:** Aseptic technique, culture/isolation, spectrophotometry, centrifugation, PCR, chromatography.
- **Research Tools:** Git, GitHub, Google Colab, ORCID, Zenodo, Preprints.org.
- **Languages (markup):** HTML5, CSS3, Markdown, LaTeX.
- **Human Languages:** Arabic (native); English (professional working proficiency; IELTS preparation in progress).

TEACHING, LEADERSHIP & INTERPERSONAL SKILLS

- Classroom management for large groups (35–50 students) and differentiated instruction
- Simplifying complex technical and scientific concepts for non-specialist audiences
- AI-assisted lesson design, visual/video-based teaching, and student psychological support
- Direct customer service and point-of-sale operations under time pressure
- Progressive operational responsibility from individual contributor to full hypermarket floor supervision
- Early leadership and labor supervision in agricultural greenhouse contracting (multi-year, self-managed team)
- Personal reliability, discipline, and financial self-sufficiency demonstrated from secondary school through university
- Bilingual communication: Arabic (native), English (professional working proficiency)

CERTIFICATIONS & LANGUAGES

- Official Experience Certificate – Al-Matar Official Primary School (Science teaching, February – May 2026)
- Python (Basic) — HackerRank Skill Certification (ID: 67EA19A6E0EB, May 2026)

- PCB Design Essentials — ITI / Mahara-Tech (November 2025)
- Biomaterials Engineering & Nanotechnology for Cancer Treatment — NRC Symposium (April 2026)
- Languages: Arabic (native); English (professional working proficiency; IELTS preparation in progress)

RESEARCH METRICS (AS OF 11 SEPTEMBER 2026)

- 1 preprint published (Preprints.org ID: 232818)
- 1 DOI issued (Zenodo: 10.5281/ZENODO.22700296)
- 1 open-source research repository (HDAFA) + engineering repositories (ESP32 Incubator, Automated Bioreactor)
- 14 benchmark tasks × 4 systems evaluated
- 100% section coverage for Analyzer-based systems; ~85% for direct-generation baselines (+15 pp)
- HDAFA pass rate 100% on completed tasks; average retries 0.00 (validator not activated on easy set)
- Transparent limitations section documenting scope, sample size, measurement threats, and non-results

ONLINE PROFILES

ORCID: <https://orcid.org/0009-0000-4920-9333>

GitHub (profile): <https://github.com/MostafaFathi-afrotoh>

GitHub (HDAFA): <https://github.com/MostafaFathi-afrotoh/hdafa>

GitHub (ESP32 Incubator): <https://github.com/MostafaFathi-afrotoh/ESP32-Egg-Incubator>

GitHub (Bioreactor): <https://github.com/MostafaFathi-afrotoh/Automated-Bioreactor>

LinkedIn: <https://www.linkedin.com/in/mustafa-fathy-abdeltawab-438b09305>

Preprints.org: <https://www.preprints.org/manuscript/202609.232818>

Zenodo DOI: <https://doi.org/10.5281/zenodo.22700296>

TARGET OPPORTUNITIES

Interdisciplinary Master's programs (Bioelectronics, Biomedical Engineering, Biological Systems Engineering) · Graduate Research positions · AI / Multi-Agent / Verifiable Systems research roles · Science / STEM Teacher or Teaching Assistant · Online Education roles · Laboratory / Quality support roles · Operations, retail, or customer-facing roles · Scholarships and funded research programs.

REFERENCES

Available upon request.

— End of Comprehensive CV Reference —

Updated 11 September 2026 · Includes HDAFA preprint release and full prior career history